

● *Original Contribution*

## QUANTITATIVE ANALYSIS OF HEPATIC MICROCIRCULATION IN RABBITS AFTER LIVER ISCHEMIA-REPERFUSION INJURY USING CONTRAST-ENHANCED ULTRASOUND

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**Abstract**—Previous studies have shown that contrast-enhanced ultrasound (CEUS) can be used quantitatively to analyze microcirculation blood perfusion in hepatocellular carcinoma patients. However, limited data have described the application of CEUS in hepatic microcirculation after liver ischemic-reperfusion injury (IRI). The purpose of this study was to explore the use of CEUS quantitatively to assess liver microcirculation after liver IRI. We randomly sorted 45 New Zealand rabbits into 3 groups (15 in each). Group A was a control group in which the rabbits underwent laparotomy alone. In groups B and C, hepatic blood was blocked for 30 min. Simultaneously, rabbits in group C underwent left lateral lobe resection. After 30 min of ischemia, CEUS was conducted after 0 h, 1 h, 6 h and 24 h of reperfusion in the 3 groups. Time-intensity curves (TICs) for CEUS were constructed and quantitative parameters (maximum intensity [IMAX], rise time [RT], time to peak [TTP] and mean transit time [mTT]) were obtained. In addition, serum aspartate aminotransferase (AST) and alanine aminotransferase (ALT) levels were computed to estimate liver function before the operation and at 0 h, 1 h, 6 h and 24 h after reperfusion, respectively. Pathologic changes in the liver after reperfusion were also observed. Simultaneously, the correlations between serum transaminase and a variety of quantitative analysis parameters were analyzed. In groups B and C, the IMAX value decreased; whereas RT, TTP, mTT and serum ALT and AST levels increased significantly in comparison with those in group A after 0 h and 1 h of reperfusion. The pathology revealed that erythrocytes were destroyed and microcirculation was disturbed. Then, at 6 h of reperfusion, the IMAX continued to decrease. Additionally, the levels of RT, TTP, mTT and serum ALT and AST increased in comparison with those at 1 h of reperfusion. The pathologic analysis revealed inflammatory cell aggregation and leukocyte infiltration. After 24 h of reperfusion, the IMAX was reduced in comparison with that of the 6-h group. The levels of RT, TTP, mTT and serum ALT and serum AST were increased in comparison with that of the 6-h group. These findings were in accordance with the pathologic analysis. In addition, serum transaminase had a negative correlation with IMAX ( $p < 0.001$ ) and a positive correlation with RT, TTP and mTT (all  $p < 0.001$ ). So, in conclusion, the quantitative analysis of CEUS can be used to assess hepatic microcirculation after liver IRI. (E-mail: [yanghonggx@163.com](mailto:yanghonggx@163.com)) © 2017 The Authors. Published by Elsevier Inc. on behalf of World Federation for Ultrasound in Medicine & Biology. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

**Key Words:** Contrast-enhanced ultrasound, Ischemia-reperfusion injury, Microcirculation dysfunction, Perfusion quantification.

### INTRODUCTION

Hepatic ischemic-reperfusion injury (IRI) is commonly observed in clinical practices, including hepatectomy, liver transplantation and hypovolemic shock. Hepatectomy and

liver transplantation have been extensively applied in liver diseases. However, IRI is an unavoidable complication of hepatectomy and liver transplantation with the consequence of ischemic insult after the restoration of blood perfusion (Jaeschke and Woolbright 2012). IRI is an important reason for liver dysfunction and failure after hepatectomy and transplantation (Saidi and Kenari 2014; Xiao et al. 2015). Hence, minimizing IRI is of great significance in the clinic. The pathology of IRI includes

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oxidative stress and the inflammatory cascade (van Golen et al. 2012a, 2012b). Two phases have been associated with liver IRI. In the early stage (first 2 h), liver damage results from oxidative stress generated by Kupffer cells (KCs), with damage to sinusoidal endothelial cells (SECs) (Samarasinghe et al. 2000; Teoh and Farrell 2003; Vollmar and Menger 2009). Then, the late stage (6–24 h) is mediated by the hepatic recruitment of macrophages and leukocytes that adhere to SECs, which secrete chemokines and express adhesion molecules (Teoh et al. 2004; Vollmar and Menger 2009) that respond to oxidative stress, hypoxia and tumor necrosis factor- $\alpha$  (TNF) (Esch et al. 2010). In conjunction, hepatic microcirculation may be injured after IRI.

Monitoring and evaluation of IRI are mainly performed by ultrasound, computed tomography (CT), magnetic resonance imaging (MRI) and pathologic assessment. Liver pathologic examination is considered the gold standard for liver IRI diagnosis. However, liver biopsy is invasive and may cause some potential complications, such as hemorrhage and embolization. CT has radiation effects, and MRI is cost prohibitive. In addition, CT and MRI are inconvenient for ill patients at the bedside. Thus, it is extremely important to explore an accurate, safe and effective method to assess liver IRI. In recent years, numerous animal experimental reports have indicated that MR and CT reflect the degree of liver IRI by monitoring hepatic microcirculation perfusion; thus, it is important to develop a non-invasive technology to monitor liver IRI (Demir et al. 2007; Guo et al. 2011; Shimizu et al. 1999; Wu, et al. 2009). Conventional color Doppler sonography has become the primary choice of imaging examination for patients in intensive care because it is flexible, convenient and viable for bedside examination. Given the low resolution of conventional ultrasound and the limitation of low-velocity blood flow, the application of color Doppler sonography in liver diseases is limited. Contrast-enhanced ultrasound (CEUS) can be used to estimate liver microcirculation in liver cancer (Tian and Wang 2016), liver fibrosis (Shin et al. 2015) and focal liver lesions (Mulé, et al. 2011). Further, study also confirmed CEUS can act as a non-invasive method to monitor liver IRI in rat models (Zhang et al. 2016). Another study showed CEUS quantitative analysis can be applied to evaluate the development of renal IRI in rabbits (Li et al. 2015). However, minimal data exist on the application of CEUS in hepatic microcirculation after liver IRI.

In this study, a rabbit model of liver IRI was established and CEUS was used to investigate changes in the time-intensity curve (TIC) parameters. In addition, serum alanine aminotransferase (ALT) and aspartate aminotransferase (AST) were used to estimate hepatic function. We examined possible associations with TIC parameters to investigate the effectiveness of CEUS in combination

with TIC in the assessment of hepatic microcirculation after rabbit liver IRI.

## MATERIALS AND METHODS

### *Establishment of animal models*

This study was performed in strict accordance with the recommendations in the Guide for the Care and Use of Laboratory Animals of the National Institutes of Health. Guangxi Medical University Laboratory Animal Centre provided 45 healthy adult New Zealand white rabbits weighing 2.3–3.0 kg (mean weight, 2.5 kg). All animals were randomly sorted into groups A, B or C ( $n = 15$  per group). Before surgery, the rabbits underwent pre-operative banned feeding for 12 h. In addition, the rabbits' fur was shaved in the region from 2 cm above the xiphoid to the abdomen, both sides to the posterior axillary line. A nearly 10-cm abdominal vertical incision was opened from the xiphoid. Subsequently, to minimize their suffering, the rabbits were anesthetized with 2.0% sodium pentobarbital (2 mL/kg) (Merck KGaA, Darmstadt, Germany) through the ear vein, and heparin (Shanghai No.1 Biochemical & Pharmaceutical Co, Ltd, Shanghai, China) (100 U) was used to prevent thrombus. Then, the abdominal skin region was sterilized with alcohol-soaked cotton balls. In group A, open abdominal surgery was performed. In this group, the liver and the duodenum ligament were separated and abdomen was closed after 30 min. In group B, open abdominal surgery was first conducted and the liver and the duodenum ligament were separated. The portal blood flow was then clamped with non-invasive arterial clips for 30 min. The color of the rabbit liver rapidly changed from bright red to dark red, indicating successful induction of ischemia. In group C, the first and second steps were the same as those in group B. The third step in group C involved the resection of the left lateral lobe, which was less than 1/4 of the entire liver. After 30 min of ischemic interval time, the arterial clips were loosened. The color of the liver changed from dark red to bright red, indicating successful reperfusion. All animals received CEUS examinations before and after 0 h, 1 h, 6 h and 24 h of reperfusion. In the sham operation group, all rabbits survived after 24 h. In the reperfusion group, 1 rabbit died before reperfusion 6 h and 4 rabbits died before reperfusion 24 h. The animals awoke about 8 h after having been anesthetized and were re-anesthetized for 24 h imaging and the operation. No rabbits were given painkillers when they were awake.

### *CEUS and imaging analysis*

CEUS was performed using the GE Logiq E9 Ultrasound System (GE Healthcare, Wauwatosa, WI, USA), with a transducer frequency of 9 MHz. All parameters were identical for each animal. The sliding control blocks for the time gain compensation curve were placed

at the same position, with a gain of 6%, a depth of 5 cm and a mechanical index of 0.16, including the dynamic range and imaging frame rate values. SonoVue (Bracco, Milan, Italy) was used as the contrast agent. Approximately 5 mL of physiologic saline was added to lyophilize SonoVue (Bracco) and evenly shaken. Then, 0.3 mL of contrast agent was rapidly injected *via* the marginal ear vein of the rabbit. Afterward, 3 mL of physiologic saline was infused into the rabbit. During the experiment, all apparatus parameters were set at fixed values. When the contrast agent was injected, the contrast program was simultaneously run, and the image capture mode was initiated. The section of the liver was captured and 1.5 min of continuous dynamic images were obtained and stored. The image file format was digital imaging and communications in medicine (DICOM). The image data were downloaded offline for analysis with SonoLiver TomTec's Image-Arena (TomTec Imaging Systems GmbH, Munich, Germany) (Rognin et al. 2010). To reduce the variation, the region of interest (ROI) was drawn by 1 investigator (Haiyuan Li). Time origin of quantitative analysis was defined as the start time of renal parenchyma enhancement, which was 3–6 s after the injection of SonoVue (Bracco). The TICs of CEUS were created and the following quantitative parameters were obtained (Fig. 1): maximum intensity (IMAX); the IMAX is relative to the percentage of the kidney, the kidney IMAX is defined as 100%; the rise time (RT), which is the time required for the contrast agent to reach the lesion; the time to peak (TTP), which is the time required for the lesions to exhibit maximum contrast intensity; and the mean transit time (mTT), which is the time required for the lesion to exhibit half peak wash-out intensity.

### Serum transaminase expression

Blood samples were collected from the femoral vein before ischemia and at 0 h, 1 h, 6 h and 24 h of reperfusion. Serum ALT and AST levels were measured by standard spectrophotometry using an automated clinical chemistry analyzer (Hitachi 7600-120, Hitachi Medical Corporation, Tokyo, Japan). The levels of serum AST and ALT can be used to estimate the degree of liver function loss.

### Pathologic analysis

Liver tissue was fixed in 10% neutral-buffered formalin at room temperature (22–28°C), embedded in paraffin, sectioned and routinely stained with hematoxylin-eosin. A pathologic examination was conducted to observe the changes in the liver tissues with a light microscope (CX31 microscope, Olympus, Tokyo, Japan).

### Statistical analysis

SPSS 19.0 (SPSS Inc., Chicago, IL, USA) statistics software was applied for the statistical analysis. All measurement data are expressed as the mean  $\pm$  standard deviation (SD). One-way ANOVA analysis was used to detect the differences between groups. Pearson correlation analysis was used to study the correlations between the serum transaminase (ALT, AST) and CEUS quantitative analysis parameters.  $p < 0.05$  was considered statistically significant.

## RESULTS

### CEUS examination results at various time points

Table 1 presented the results of the quantitative analysis parameters in the three groups at various time points.

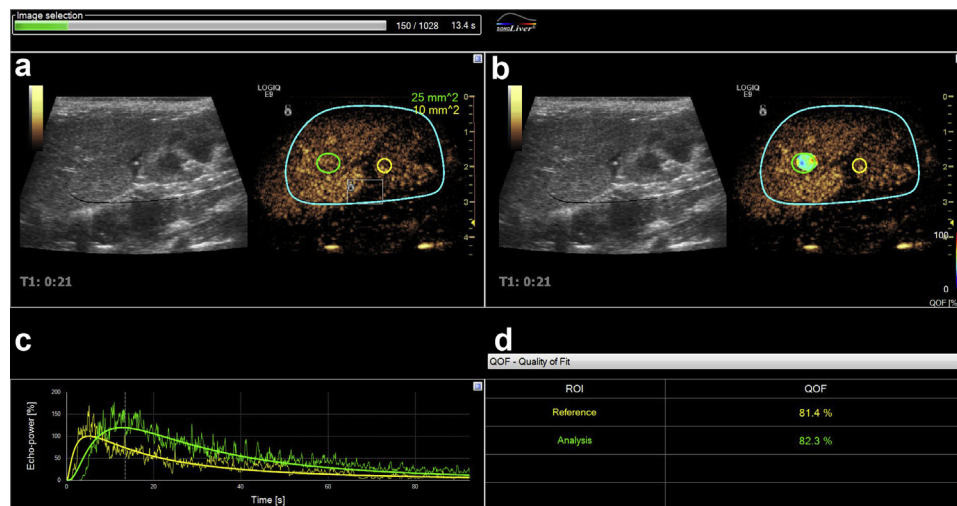


Fig. 1. Quantitative analysis of liver IRI. The parallel B-scan, the CEUS mode and the time intensity curve analysis were generated by SonoLiver TomTec's Image-Arena (TomTec Imaging Systems GmbH, Munich, Germany). IRI = ischemic-reperfusion injury; CEUS = contrast-enhanced ultrasound.

Table 1. The results of quantitative analysis parameters in three groups at various time points

| Group | Before IRI    | IRI 0 h        | IRI 1 h        | IRI 6 h        | IRI 24 h        | <i>p</i> |
|-------|---------------|----------------|----------------|----------------|-----------------|----------|
| IMAX  |               |                |                |                |                 |          |
| A     | 73.64 ± 35.53 | 74.08 ± 40.19  | 81.10 ± 25.61  | 70.41 ± 40.25  | 76.47 ± 10.45   | >0.05    |
| B     | 75.08 ± 30.42 | 69.01 ± 30.42  | 66.75 ± 35.40  | 44.36 ± 20.12  | 30.20 ± 18.50   | <0.05    |
| C     | 70.41 ± 23.25 | 67.41 ± 23.32  | 65.32 ± 30.54  | 43.44 ± 23.30  | 29.90 ± 18.25   | <0.05    |
| RT    |               |                |                |                |                 |          |
| A     | 12.48 ± 5.64  | 13.55 ± 5.48   | 13.98 ± 5.30   | 12.11 ± 4.46   | 11.54 ± 5.50    | >0.05    |
| B     | 13.21 ± 5.30  | 16.36 ± 4.53   | 17.30 ± 5.44   | 22.43 ± 8.76   | 23.47 ± 17.20   | <0.05    |
| C     | 13.01 ± 5.3   | 17.42 ± 4.75   | 17.55 ± 7.20   | 21.45 ± 10.25  | 23.80 ± 19.25   | <0.05    |
| TTP   |               |                |                |                |                 |          |
| A     | 13.53 ± 5.5   | 14.0 ± 6.75    | 14.35 ± 4.41   | 12.98 ± 7.25   | 14.32 ± 7.20    | >0.05    |
| B     | 14.25 ± 6.5   | 17.69 ± 10.32  | 18.82 ± 11.33  | 22.67 ± 17.44  | 24.44 ± 19.25   | <0.05    |
| C     | 14.90 ± 3.7   | 18.42 ± 9.80   | 19.90 ± 10.25  | 23.59 ± 12.53  | 25.12 ± 12.11   | <0.05    |
| mTT   |               |                |                |                |                 |          |
| A     | 103.8 ± 75.3  | 100.64 ± 70.2  | 92.66 ± 3.01   | 106.33 ± 46.56 | 116.7 ± 78.20   | >0.05    |
| B     | 90.66 ± 3.01  | 94.32 ± 47.50  | 90.47 ± 7.45   | 140.37 ± 97.42 | 170.55 ± 120.55 | <0.05    |
| C     | 81.40 ± 35.32 | 110.37 ± 42.05 | 100.67 ± 49.05 | 147.84 ± 70.42 | 181.92 ± 35.32  | <0.05    |

IRI = ischemic-reperfusion injury; IMAX = maximum intensity; RT = rising time; TTP = time to peak; mTT = mean transit time.

The CEUS quantitative analysis parameters (IMAX, RT, TTP, mTT) in groups A, B and C were not statistically significant ( $p > 0.05$ ) before ischemia. After reperfusion, the IMAX value gradually decreased. The RT, TTP and mTT values increased in groups B and C compared with those of group A (Fig. 2). However, the parameters showed no significant difference between groups B and C at any time after reperfusion ( $p > 0.05$ ). After reperfusion, the IMAX values in groups B and C were strikingly reduced compared with those of group A at different time

points; whereas the RT, TTP and mTT in groups B and C were all significantly increased compared with those in group A ( $p < 0.05$ ) (Fig. 2).

#### Serum transaminase values at various time points

Table 2 presents the serum ALT and AST values at various time points. In group A, serum ALT exhibited no significant difference at the various time points ( $p > 0.05$ ). In group B, the ALT level at 0 h, 1 h, 6 h and 24 h of reperfusion were  $105.8 \pm 25.3$  U/L,  $130.9 \pm 20.6$  U/L,  $180.6 \pm 24.6$  U/L

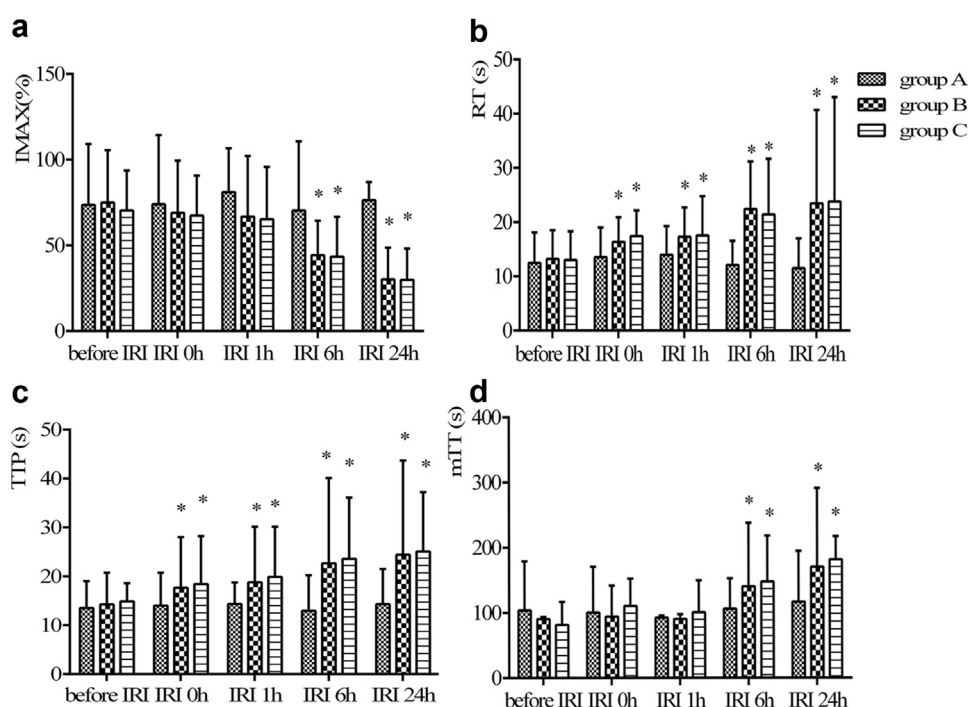


Fig. 2. The CEUS quantitative analysis parameters at various time points in the three groups (a) IMAX; (b) RT; (c) TTP; and (d) mTT. \* $p < 0.05$ . CEUS = contrast-enhanced ultrasound; IMAX = maximum intensity; RT = rise time; TTP = time to peak; mTT = mean transit time.



Table 2. The results of serum transaminase at various time points (U/L)

| Group      | A           |             | B            |              | C            |              |
|------------|-------------|-------------|--------------|--------------|--------------|--------------|
|            | ALT         | AST         | ALT          | AST          | ALT          | AST          |
| before IRI | 39.4 ± 21.2 | 44.4 ± 19.1 | 40.3 ± 15.2  | 43.6 ± 20.8  | 44.2 ± 15.2  | 43.5 ± 20.8  |
| IRI 0 h    | 37.3 ± 15.3 | 48.0 ± 20.8 | 105.8 ± 25.3 | 160.8 ± 25.8 | 109.6 ± 13.2 | 187.0 ± 15.1 |
| IRI 1 h    | 42.6 ± 13.2 | 50.6 ± 18.6 | 130.9 ± 20.6 | 209.0 ± 30.9 | 142.0 ± 18.2 | 211.7 ± 25.6 |
| IRI 6 h    | 40.8 ± 10.5 | 46.5 ± 20.1 | 180.6 ± 24.6 | 459.3 ± 28.6 | 193.3 ± 39.4 | 475.3 ± 22.3 |
| IRI 24 h   | 45.8 ± 18.2 | 43.6 ± 18.4 | 280.6 ± 30.8 | 885.6 ± 30.9 | 302.1 ± 29.9 | 903.3 ± 28.3 |
| <i>p</i>   | >0.05       | >0.05       | <0.05        | <0.05        | <0.05        | <0.05        |

ALT = alanine aminotransferase; AST = aspartate aminotransferase; IRI = ischemic-reperfusion injury.

and  $280.6 \pm 30.8$  U/L, respectively. The ALT levels in this group were considerably increased after reperfusion (Fig. 3a). In group C, the serum ALT levels were  $109.6 \pm 13.2$  U/L after 0 h reperfusion,  $142.0 \pm 18.2$  U/L after 1 h reperfusion,  $193.3 \pm 39.4$  U/L after 6 h reperfusion and  $302.1 \pm 29.9$  U/L after 24 h reperfusion. The ALT values in group C were significantly different at the various time points ( $p < 0.05$ ) (Fig. 3a). However, no significant difference was found in the change in the serum ALT levels between groups B and C ( $p > 0.05$ ) at the various time points after reperfusion. After reperfusion, the serum ALT levels in groups B and C were significantly increased compared with those in group A at the various time points (Fig. 3a).

The serum AST levels in group A exhibited no difference before and after reperfusion ( $p > 0.05$ ). In group B, the serum AST levels were  $43.6 \pm 20.8$  U/L before ischemia,  $160.8 \pm 25.8$  after 0 h of reperfusion,  $209.0 \pm 30.9$  U/L after 1 h of reperfusion,  $459.3 \pm 28.6$  U/L after 6 h of reperfusion and  $885.6 \pm 30.9$  U/L after 24 h of reperfusion, with significant differences observed at the various times (all  $p < 0.05$ ) (Fig. 3b). In group C, the AST levels at 0 h, 1 h, 6 h and 24 h of reperfusion were  $187.0 \pm 15.1$  U/L,  $211.7 \pm 25.6$  U/L,  $475.3 \pm 22.3$  U/L, and  $903.3 \pm 28.3$  U/L, respectively. Statistically significant differences were noted in this group (Fig. 3b). However, no differences were found in groups B and group C after reperfusion ( $p > 0.05$ ). In groups B and C, the serum AST levels were increased compared with those of group A at the various time points (all  $p < 0.05$ ) (Fig. 3b).

#### Correlation between quantitative analysis parameters and serum transaminase

Table 3 presents the correlation between the quantitative analysis parameters and transaminase. In group B, the Spearman correlation test revealed positive correlations between serum ALT and the following quantitative analysis parameters: IMAX ( $r = -0.490$ ,  $p < 0.001$ ), RT ( $r = 0.494$ ,  $p < 0.001$ ), TTP ( $r = 0.284$ ,  $p < 0.001$ ) and mTT ( $r = 0.561$ ,  $p < 0.001$ ). In this group, positive correlations between serum AST and the following quantitative analysis parameters were observed: IMAX ( $r = -0.529$ ,  $p < 0.001$ ), RT ( $r = 0.389$ ,  $p < 0.001$ ), TTP ( $r = 0.298$ ,  $p < 0.001$ ) and mTT ( $r = 0.574$ ,  $p < 0.001$ ). In group C, the Spearman correlation test revealed positive correlations between serum ALT and the following quantitative analysis parameters: IMAX ( $r = -0.518$ ,  $p < 0.001$ ), RT ( $r = 0.385$ ,  $p < 0.001$ ), TTP ( $r = 0.394$ ,  $p < 0.001$ ) and mTT ( $r = 0.576$ ,  $p < 0.001$ ). In this group, positive correlations between serum AST and the following quantitative analysis parameters were observed: IMAX ( $r = -0.549$ ,  $p < 0.001$ ), RT ( $r = 0.382$ ,  $p < 0.001$ ), TTP ( $r = 0.443$ ,  $p < 0.001$ ) and mTT ( $r = 0.555$ ,  $p < 0.001$ ).

#### Pathologic examination results at various time points

After 0 h of reperfusion, the pathologic examination revealed hepatocellular swelling, follicular fatty changes, thickening of the hepatic plate and narrowing of the sinus (Fig. 4a). The liver injury manifested as an aggregation of red cells in hepatic sinusoids and an obstruction of the vessel lumen after 1 h of reperfusion (Fig. 4b).

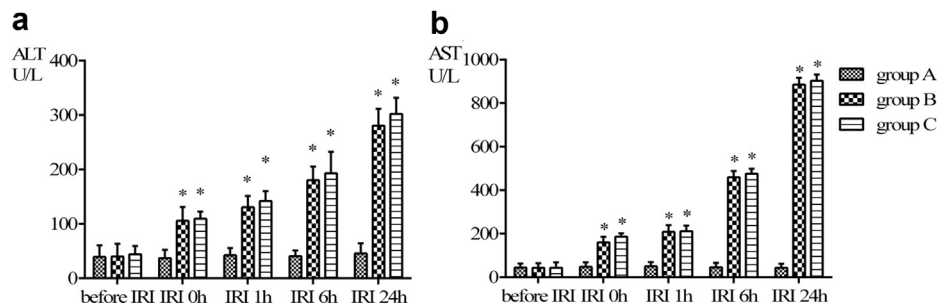


Fig. 3. Serum transaminase values at various time points in groups B and C. (a) ALT levels at various time points; (b) AST levels at various time points. \* $p < 0.05$ . ALT = alanine aminotransferase; AST = aspartate aminotransferase.

Table 3. Correlation between quantitative analysis parameters and serum transaminase (U/L)

| Parameter | Group A |        | Group B |        | Group C |        |
|-----------|---------|--------|---------|--------|---------|--------|
|           | ALT     | AST    | ALT     | AST    | ALT     | AST    |
| IMAX      |         |        |         |        |         |        |
| R         | 0.207   | 0.026  | -0.490  | -0.529 | -0.518  | -0.549 |
| P         | 0.110   | 0.753  | 0.000   | 0.000  | 0.000   | 0.000  |
| RT        |         |        |         |        |         |        |
| R         | 0.014   | 0.062  | 0.494   | 0.389  | 0.385   | 0.382  |
| P         | 0.861   | 0.449  | 0.000   | 0.000  | 0.000   | 0.000  |
| TTP       |         |        |         |        |         |        |
| R         | 0.017   | -0.034 | 0.284   | 0.298  | 0.394   | 0.443  |
| P         | 0.838   | 0.679  | 0.000   | 0.000  | 0.000   | 0.000  |
| mTT       |         |        |         |        |         |        |
| R         | 0.002   | 0.105  | 0.561   | 0.574  | 0.576   | 0.555  |
| P         | 0.980   | 0.199  | 0.000   | 0.000  | 0.000   | 0.000  |

IMAX = maximum intensity; RT = rising time; TTP = time to peak; mTT = mean transit time.

After 6 h of reperfusion, the injury manifested as aggravation of liver cell edema, ballooning degeneration and neutral granulocyte infiltration (Fig. 4c). After 24 h of reperfusion, pathologic changes manifested as hepatic cell atrophy, necrosis and hepatic sinus collapse (Fig. 4d).

## DISCUSSION

IRI occurs when the blood is interrupted and then restored. Damage caused by IRI is irreversible for the organ. IRI is an inevitable part of liver surgery, such as liver transplantation and resection (Busuttil and Tanaka 2003; Weigand et al. 2012). During hepatic resections, vascular occlusion of liver hilum is the cause of IRI (Weigand et al. 2012). During transplantation, cold storage of liver graft, warm reperfusion and implantation also cause damage (Casillas-Ramirez et al. 2006). At the cellular level, ischemia can cause adenosine triphosphate breakdown. This phenomenon also results in the accumulation of hypoxanthine, mitochondrial de-energization and ionic alteration, leading to liver cell necrosis (Pasut et al. 2016). Neutrophils are the main responders and the inflammatory response and the severity are related to IRI (Schofield et al. 2013). During reperfusion, the generation of reactive oxygen species (ROS) promotes mitochondrial oxidative stress and inflammatory mediators, which contribute to damage and cell death (Casillas-Ramirez et al. 2006).

Although CT and MR can be applied to assess liver IRI and microcirculation, CT has radiation effects, and MR is costly. Moreover, CT and MR are not available for

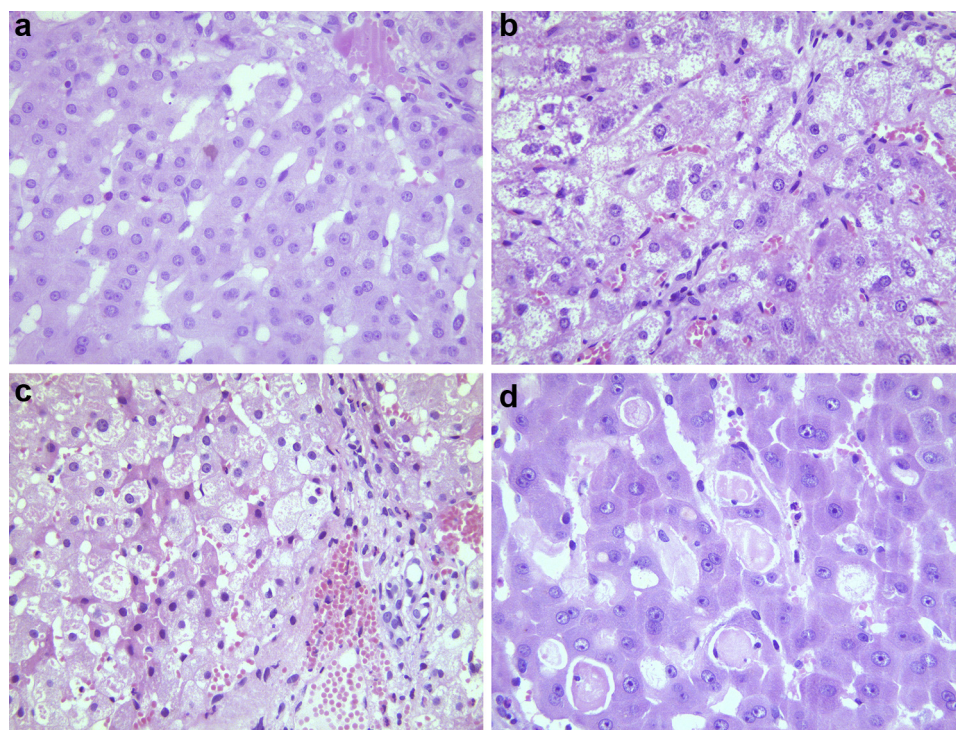


Fig. 4. Pathologic changes at various time points after liver IRI (a) The pathologic changes included hepatocellular swelling, follicular fatty changes, thickening of the hepatic plate and narrowing of the sinus after 0 h of reperfusion. (b) The pathologic changes included the aggregation of red cells in the hepatic sinusoids and the obstruction of the vessel lumen after 1 h of reperfusion. (c) The pathologic changes included the aggravation of liver cell edema, ballooning degeneration and neutral granulocyte infiltration after 6 h of reperfusion. (d) The pathologic changes included hepatic cell atrophy, necrosis and hepatic sinus collapse after 24 h of reperfusion. The original magnification of a–b was 400×. IRI = ischemic-reperfusion injury.

ill patients at the bedside. However, CEUS is a non-invasive, non-radioactive, cost-effective imaging technique that is available at the bedside. Furthermore, CEUS is a real-time angiographic-like imaging technique and has a high diagnostic efficiency, which can be used to monitor the microcirculation of the liver graft, facilitating the visualization of blood vessels (Claudon et al. 2008). In addition, CEUS, a safe and effective imaging procedure, is widely applied in clinical practice. The incidence of severe adverse reactions to CEUS is 0.0086%–0.012% (Kusnetzky et al. 2008, Piscaglia and Bolondi 2006). A report showed CEUS can help to monitor renal IRI in animal models (Li et al. 2015). Moreover, CEUS can be applied to patients with renal dysfunction because the gas is metabolized by the respiratory system (Jakobsen et al. 2005). TIC is obtained from the CEUS dynamic change information, using quantitative analysis software. Wei et al. (2001) demonstrated that the quantitative parameters obtained from TIC curves correlated with changes in blood flow.

In this study, RT, TTP and mTT were higher in groups B and C than in group A, indicating that blood flow was reduced. After perfusion, RT, TTP and mTT in groups B and C were increased, indicating that microcirculation dysfunction occurred. This finding was confirmed by pathologic analysis. After 1 h of reperfusion, pathologic changes were maintained as the narrowing of hepatic sinus and hepatic cells swelling. After 6 h of reperfusion, RT, TTP and mTT were increased compared with the results at 1 h, and the pathologic examination showed disappearance of hepatic sinus and liver injury. After 24 h of reperfusion, hepatic perfusion was further reduced. Simultaneously, pathologic changes included hepatic cell atrophy, necrosis and hepatic sinus collapse. IMAX is a parameter that reflects blood volume. After reperfusion, the values of IMAX in groups B and C were lower than those in group A, indicating that hepatic microcirculation worsened.

Serum ALT and AST are important indices for monitoring hepatic injury (Jin et al. 2013). In our study, serum ALT and AST levels dramatically increased after 1 h of reperfusion. The results were consistent with pathologic changes.

The results in this study were similar to the finding demonstrated by Zhang et al. (2016). Moreover, rabbit models of liver IRI in groups B and C were mimicked clinical procedures. And our results showed that serum ALT and AST had correlations with quantitative analysis parameters. These were consistent with the pathologic changes. Multiple factors are associated with IRI. A potential reason for IRI is the imbalance of nitric oxide (NO) and endothelin (ET). ET is a peptide substance secreted by vascular endothelial cells that has an impact on vascular contraction. During ischemia, hypoxia can promote endothelial cells to induce more ET (Stewart et al. 2014). On

the other hand, the accumulation of intestinal toxins in the liver also increases ET levels. However, NO acts as a main antagonist of NO and is down-regulation during ischemia (Toshihiro et al. 2015). ET can act on hepatic sinusoids, resulting in the stenosis of the liver sinusoid. In addition, hepatic and endothelial cells are located in a state of hypoxia, and microvilli membrane swell, causing slurry bubble formation. Liver cell edema expands into the hepatic blood sinus. In addition, blood cells accumulate in the hepatic sinusoid. When reperfusion occurs, the sinus becomes narrow, resulting in the microcirculation disorder (Peralta et al. 2013).

## CONCLUSION

CEUS can be used to monitor microcirculation dysfunction after liver IRI, which corresponds to changes in pathology and serum enzymes. CEUS may reflect the pathologic changes in the liver and quantitative analysis can be used to measure hepatic microcirculation after liver IRI. CEUS is a non-invasive imaging procedure that can be applied to measure IRI in the early stage, providing a feasible method for exploring the pathogenesis, diagnosis and treatment of IRI in a clinical setting.

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